

01-ORM相关模版代码

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安装SQLAlchemy



Python |

```
1 pip install "sqlalchemy[asyncio]" aiomysql
```

ORM 配置

```
1  from sqlalchemy.ext.asyncio import async_sessionmaker, AsyncSession, create_async_engine
2
3  # 数据库URL
4  ASYNC_DATABASE_URL = "mysql+aiomysql://root:123456@localhost:3306/news_app?charset=utf8mb4"
5
6  # 创建异步引擎
7  async_engine = create_async_engine(
8      ASYNC_DATABASE_URL,
9      echo=True, # 可选: 输出SQL日志
10     pool_size=10, # 设置连接池中保持的持久连接数
11     max_overflow=20 # 设置连接池允许创建的额外连接数
12 )
13
14 # 创建异步会话工厂
15 AsyncSessionLocal = async_sessionmaker(
16     bind=async_engine,
17     class_=AsyncSession,
18     expire_on_commit=False
19 )
20
21 # 依赖项, 用于获取数据库会话
22 async def get_db():
23     async with AsyncSessionLocal() as session:
24         try:
25             yield session
26             await session.commit()
27         except Exception:
28             await session.rollback()
29             raise
30         finally:
31             await session.close()
32
```

News 模型类

```
1 class News(Base):
2     __tablename__ = "news"
3
4     # 创建索引：提升查询速度
5     __table_args__ = (
6         Index('fk_news_category_idx', 'category_id'),
7         Index('idx_publish_time', 'publish_time')
8     )
9
10    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True, comment="新闻ID")
11    title: Mapped[str] = mapped_column(String(255), nullable=False, comment="新闻标题")
12    description: Mapped[Optional[str]] = mapped_column(String(500), comment="新闻简介")
13    content: Mapped[str] = mapped_column(Text, nullable=False, comment="新闻内容")
14    image: Mapped[Optional[str]] = mapped_column(String(255), comment="封面图片URL")
15    author: Mapped[Optional[str]] = mapped_column(String(50), comment="作者")
16    category_id: Mapped[int] = mapped_column(Integer, ForeignKey('news_category.id'), nullable=False, comment="分类ID")
17    views: Mapped[int] = mapped_column(Integer, default=0, nullable=False, comment="浏览量")
18    publish_time: Mapped[datetime] = mapped_column(DateTime, default=datetime.now, comment="发布时间")
19
20    def __repr__(self):
21        return f"<News(id={self.id}, title='{self.title}', views={self.views})>"
```

用户表和用户 Token 表模型类

```
1  from datetime import datetime
2  from typing import Optional
3
4  from sqlalchemy import Index, Integer, String, Enum, DateTime, ForeignKey
5  from sqlalchemy.orm import DeclarativeBase, Mapped, mapped_column
6
7
8  class Base(DeclarativeBase):
9      pass
10
11
12  class User(Base):
13      """
14      用户信息表ORM模型
15      """
16      __tablename__ = 'user'
17
18      # 创建索引
19      __table_args__ = (
20          Index('username_UNIQUE', 'username'),
21          Index('phone_UNIQUE', 'phone'),
22      )
23
24      id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True, comment="用户ID")
25      username: Mapped[str] = mapped_column(String(50), unique=True, nullable=False, comment="用户名")
26      password: Mapped[str] = mapped_column(String(255), nullable=False, comment="密码 (加密存储)")
27      nickname: Mapped[Optional[str]] = mapped_column(String(50), comment="昵称")
28      avatar: Mapped[Optional[str]] = mapped_column(String(255), comment="头像URL",
29                                                    default='https://fastly.jsdelivrivr.net/npm/@vant/assets/cat.jpeg')
30      gender: Mapped[Optional[str]] = mapped_column(Enum('male', 'female', 'unknown'), comment="性别", default='unknown')
31      bio: Mapped[Optional[str]] = mapped_column(String(500), comment="个人简介", default='这个人很懒，什么都没留下')
32      phone: Mapped[Optional[str]] = mapped_column(String(20), unique=True, comment="手机号")
33      created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.now(), comment="创建时间")
34      updated_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.now(), onupdate=datetime.now(),
```

```

35                                                                 comment="更新时间")
36
37
38
39     def __repr__(self):
40         return f"<User(id={self.id}, username='{self.username}', nickname
41         ='{self.nickname}')>"
42
43     class UserToken(Base):
44         """
45         用户令牌表ORM模型
46         """
47         __tablename__ = 'user_token'
48
49         # 创建索引
50         __table_args__ = (
51             Index('token_UNIQUE', 'token'),
52             Index('fk_user_token_user_idx', 'user_id'),
53         )
54
55         id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True, comment="令牌ID")
56         user_id: Mapped[int] = mapped_column(Integer, ForeignKey(User.id), nullable=False, comment="用户ID")
57         token: Mapped[str] = mapped_column(String(255), unique=True, nullable=False, comment="令牌值")
58         expires_at: Mapped[datetime] = mapped_column(DateTime, nullable=False, comment="过期时间")
59         created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.now(), comment="创建时间")
60
61
62     def __repr__(self):
63         return f"<UserToken(id={self.id}, user_id={self.user_id}, token='{self.token}')>"

```

收藏表模型类

```
1  from datetime import datetime
2
3  from sqlalchemy import UniqueConstraint, Index, Integer, ForeignKey, DateTime
4  from sqlalchemy.orm import DeclarativeBase, Mapped, mapped_column
5
6  from models.news import News
7  from models.users import User
8
9
10 class Base(DeclarativeBase):
11     pass
12
13
14 class Favorite(Base):
15     """
16     收藏表ORM模型
17     """
18     __tablename__ = 'favorite'
19
20     # 创建索引
21     __table_args__ = (
22         UniqueConstraint('user_id', 'news_id', name='user_news_unique'),
23         Index('fk_favorite_user_idx', 'user_id'),
24         Index('fk_favorite_news_idx', 'news_id'),
25     )
26
27     id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True, comment="收藏ID")
28     user_id: Mapped[int] = mapped_column(Integer, ForeignKey(User.id), nullable=False, comment="用户ID")
29     news_id: Mapped[int] = mapped_column(Integer, ForeignKey(News.id), nullable=False, comment="新闻ID")
30     created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow, nullable=False, comment="收藏时间")
31
32     def __repr__(self):
33         return f"<Favorite(id={self.id}, user_id={self.user_id}, news_id={self.news_id}, created_at={self.created_at})>"
34
```

浏览历史表模型类

```
1 from sqlalchemy.orm import Mapped, mapped_column, DeclarativeBase
2 from sqlalchemy import Integer, DateTime, ForeignKey, Index
3 from datetime import datetime
4 from .users import User
5 from .news import News
6
7 class Base(DeclarativeBase):
8     pass
9
10
11 class History(Base):
12     """
13     浏览历史表ORM模型
14     """
15     __tablename__ = 'history'
16
17     # 创建索引
18     __table_args__ = (
19         Index('fk_history_user_idx', 'user_id'),
20         Index('fk_history_news_idx', 'news_id'),
21         Index('idx_view_time', 'view_time'),
22     )
23
24     id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True, comment="历史ID")
25     user_id: Mapped[int] = mapped_column(Integer, ForeignKey(User.id), nullable=False, comment="用户ID")
26     news_id: Mapped[int] = mapped_column(Integer, ForeignKey(News.id), nullable=False, comment="新闻ID")
27     view_time: Mapped[datetime] = mapped_column(DateTime, default=datetime.now, nullable=False, comment="浏览时间")
28
29
30 def __repr__(self):
31     return f"<History(id={self.id}, user_id={self.user_id}, news_id={self.news_id}, view_time={self.view_time})>"
32
```